

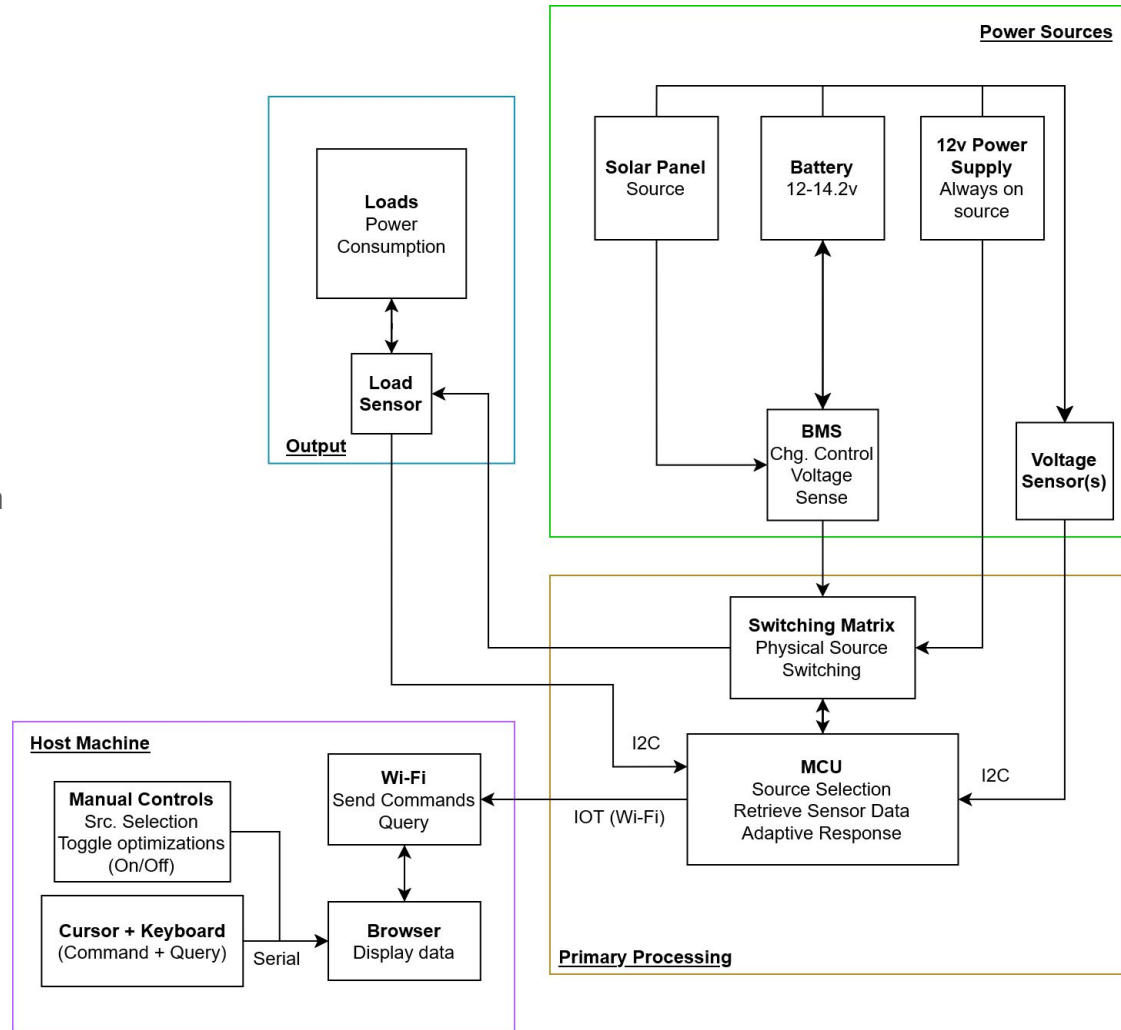
Real Time Smart Energy Management System

*Optimizes source selection between grid and renewable power sources
with real time sensing and Time of Use pricing.*

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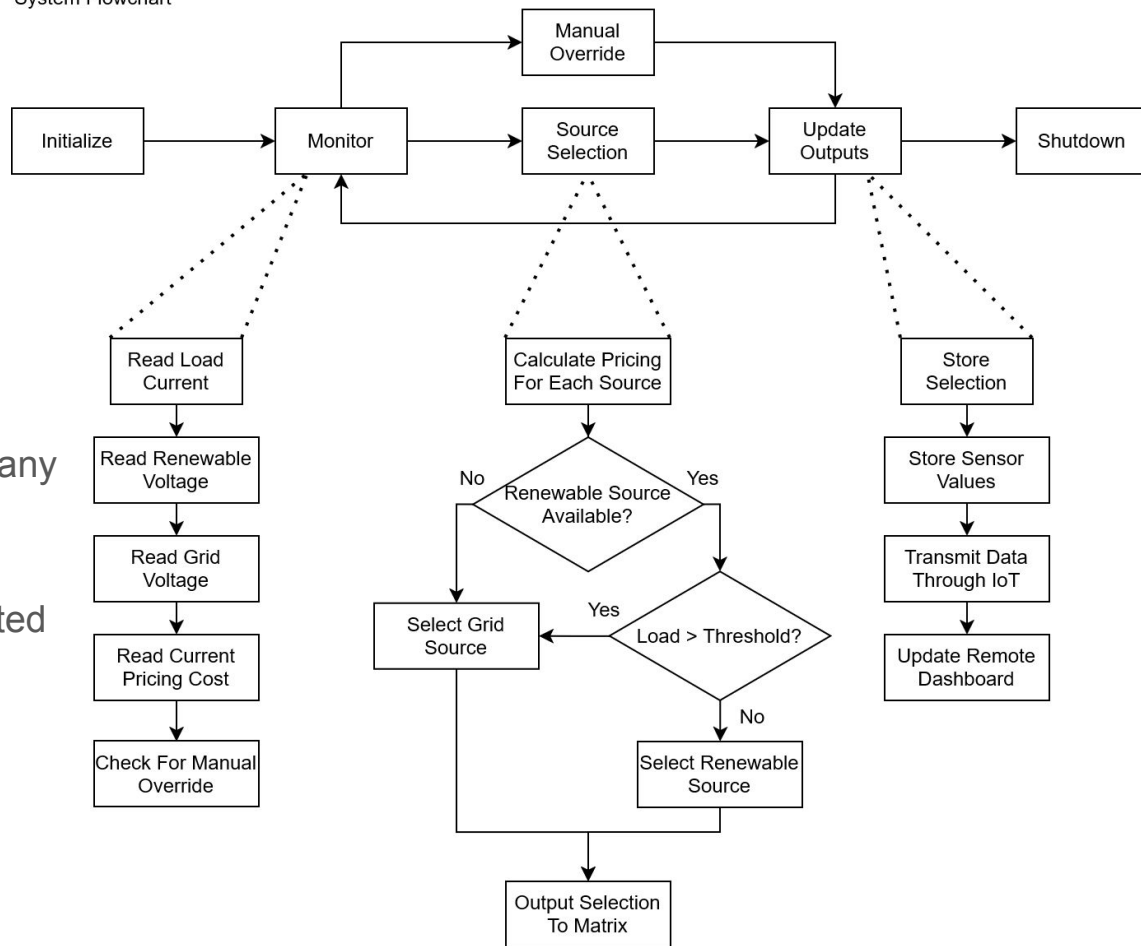
System Description

- **Power Sources:**
 - Renewable, Always on “Grid”
 - Managed renewable energy
- **Primary Processing:**
 - Microcontroller
 - Reactive source selection
 - Active data monitoring + optimization
- **Host Machine:**
 - Interactive dashboard
 - Status Display + Manual Control
- **Output:**
 - “Household” devices
 - Real time load measurements



Dynamic Source Optimization

System Flowchart



System Description

Basic Principle:

1. Gather and read information
2. Select and switch source
3. Update dashboard display

- Manual override can be activated at any time
- Costs are stored locally
- Smooth source selection (uninterrupted switches)

EMS Pseudo Code

START EMS AUTO DECISION LOOP

Every 1 second:

1. Update simulated clock

- Advance sim time based on sim speed
- Update current hour (simHour)

2. Determine current price band

IF tariffProfile == Riverside summer:
peak hours = 16 to 20
peak price = 0.55
off-peak price = 0.22
ELSE IF tariffProfile == Chicago winter:
peak hours = 7 to 9 and 17 to 19
peak price = 0.22
off-peak price = 0.16

currentPrice = price at simHour
inPeak = whether simHour is inside peak period

3. Read battery condition

batteryVoltage = measured battery voltage
batterySOC = estimated battery state of charge

4. Decide whether battery path is allowed

IF currently using GRID:
battery is allowed only if
batterySOC >= 35%
AND batteryVoltage >= 12.20 V
ELSE IF currently using BATTERY path:
battery is allowed to continue if
batterySOC >= 25%
AND batteryVoltage >= 12.00 V

battOk = result of this hysteresis check

5. Time-of-use source selection rule

IF battOk == true AND inPeak == true:
candidateRenew = true
ELSE:
candidateRenew = false

Meaning:

- During expensive peak hours, **use battery if battery is healthy**
- During off-peak hours, use grid
- If battery becomes weak, **fall back to grid**

6. Add decision confirmation delay

IF candidateRenew changed since last check:
remember new candidateRenew
store the sim-time when the change happened

IF candidateRenew has stayed unchanged for at least 5 simulated minutes:
latchedWantRenew = candidateRenew

Meaning:

- Do **not switch immediately** on every small change
- Only switch after condition stays stable for 5 simulated minutes

7. Convert decision into requested relay states

IF latchedWantRenew == true:
relay1_on = false // GRID OFF
relay2_on = true // BATTERY / RENEWABLE PATH ON
ELSE:
relay1_on = true // GRID ON
relay2_on = false // BATTERY / RENEWABLE PATH OFF

8. Apply relay switching safely

- Never allow both relays on at the same time
- Enforce minimum dwell time between switches:
manual mode: 2500 ms
auto mode: 600 ms
- If a change is needed:
 - a. turn both relays off
 - b. wait 150 ms break-before-make time
 - c. turn on only the selected relay

9. Track actual applied relay state

- Save whether GRID relay is physically on
- Save whether BATTERY relay is physically on

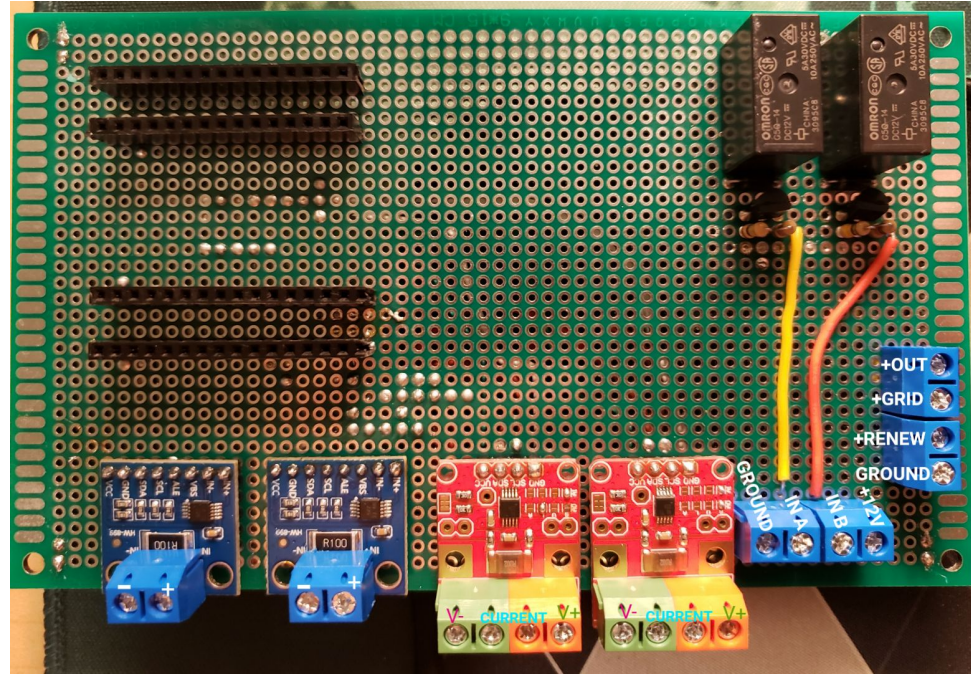
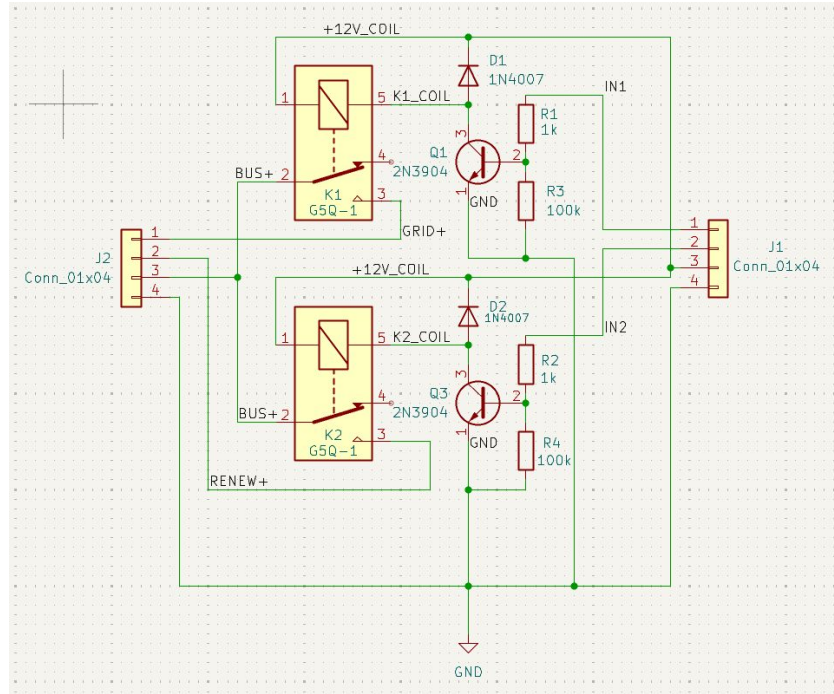
10. Final result

IF relay1 is on:
source = GRID
ELSE IF relay2 is on:
source = BATTERY / RENEWABLE PATH
ELSE:

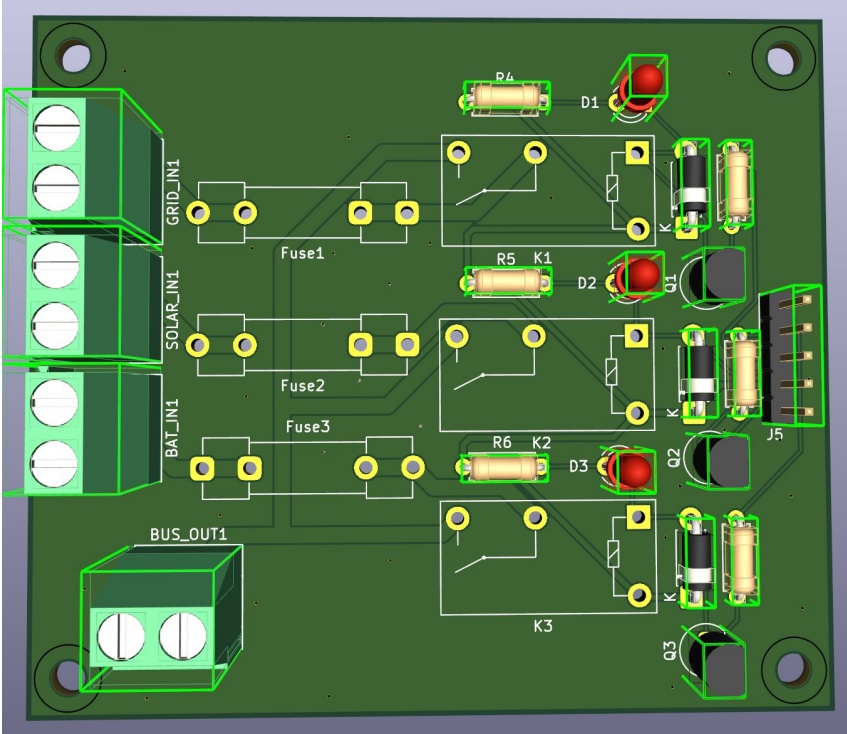
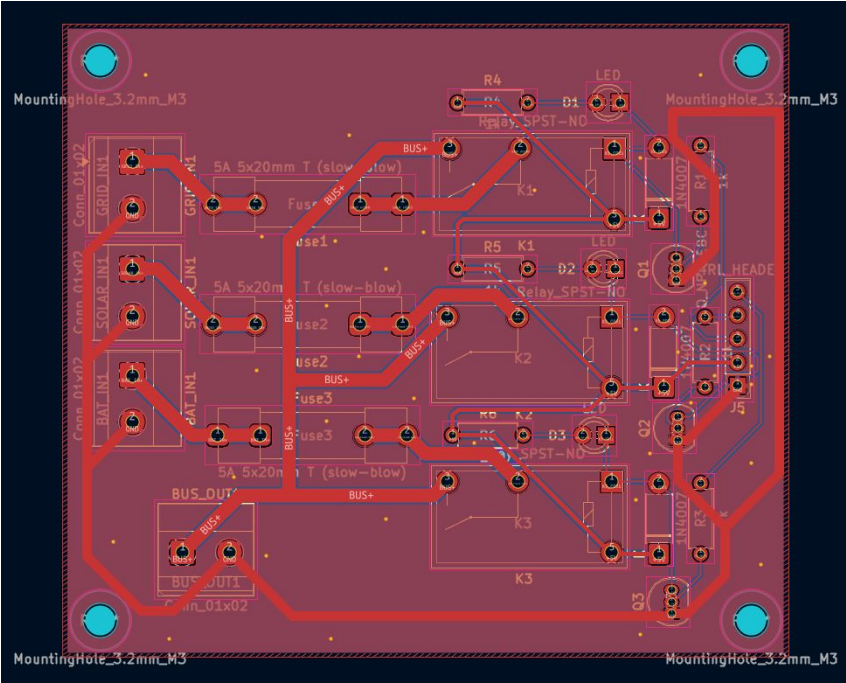
BOTTOM LINE:

```
if (cheap period) {  
    use grid  
}  
else if (expensive period) {  
    use battery  
}  
else { (unsafe/unnecessary battery use)  
    return to grid.  
}
```

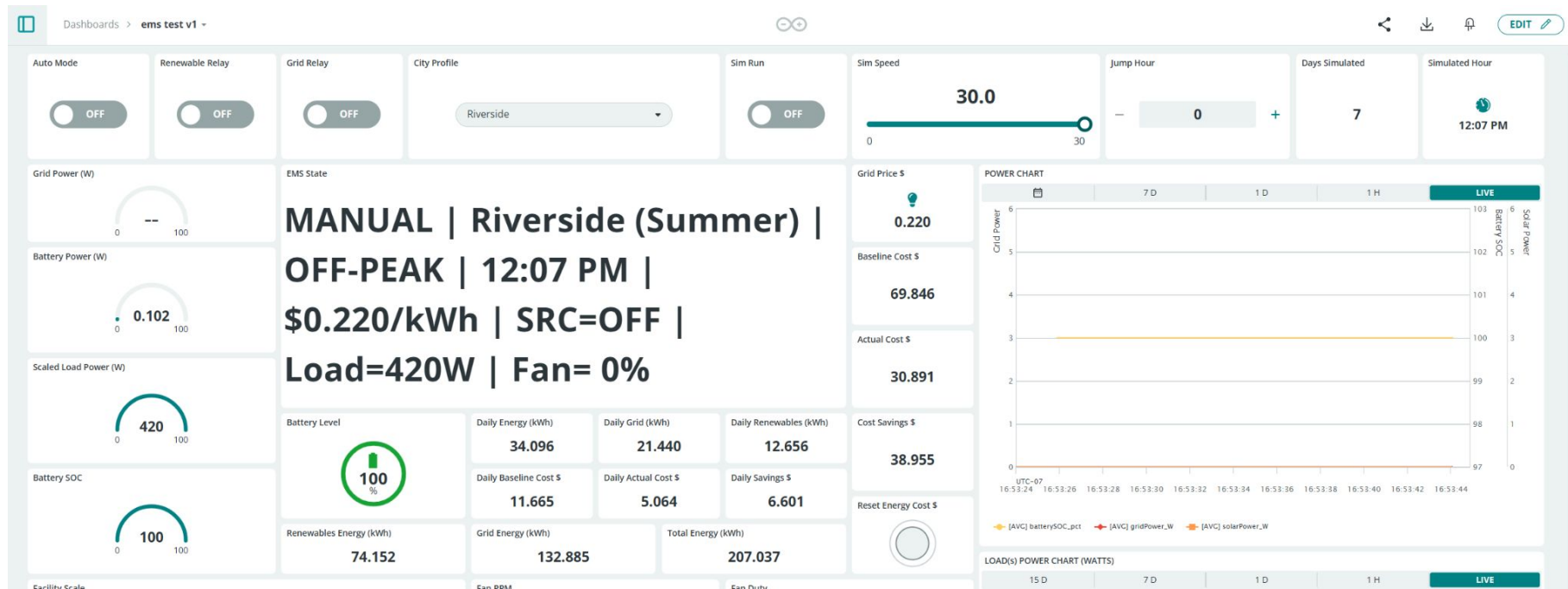
Schematic



PCB



Remote Dashboard



[Video demo](#)

Pricing Analysis + Project Cost

Relay Matrix: Relays (\$9) + PCB prototype board and terminals/headers (\$10) + 1N4007 Rectifier Diodes + 2N3904 NPN Transistors + Additional Passives = ~\$37

MCU + sensors: ESP32 (\$12) + INA226 current/power monitoring sensors (\$24) = ~\$36

Power Sources: 25 Watts 12V Off Grid Solar Battery Charger Kit (\$38) + 12V 5AH SLA Rechargeable Battery (\$18) + DC power supply (\$47) = ~\$103

Cost to Manufacture = ~\$176

